
LiverBridge

Predicting translational failure and prioritising human-relevant 3D validation for drug-induced liver injury

Concrete hackathon project report and implementation plan

One-sentence pitch. Drug candidates often fail because scalable early screens do not preserve the human biological context that determines toxicity, while organoids and organs-on-chips are too expensive to run on every compound. LiverBridge predicts where a conventional screen is likely to fail, explains the human-relevant liability, and recommends which compounds should be escalated to 3D Liver-Chip validation.

Event	London AI × Bio Hackathon
Working name	LiverBridge (team may rename)
Primary use case	Drug-induced liver injury (DILI)
Build horizon	24–48 hour prototype
Core technologies	Molecular representation, toxicogenomics, exposure-aware ML, scientific reasoning agent
Date	18 September 2026

1 Executive summary

Question. Can we identify drug candidates for which a cheap conventional screen underestimates human liver toxicity, and therefore decide which candidates warrant an expensive human-relevant 3D experiment?

Proposed answer. Build a focused decision-support prototype for DILI. It compares a structure-only baseline with a richer human-biology model using molecular representation, primary-human-hepatocyte transcriptomic or pathway responses, and clinical exposure. The system reports (i) human DILI risk, (ii) a calibrated translational-discordance score, (iii) uncertainty, (iv) likely mechanisms, and (v) a recommendation to escalate or not escalate the compound to 3D Liver-Chip testing.

Why this can be prize-level. The project does not merely predict toxicity. Its distinctive claim is that it predicts *the failure of the preclinical model* and converts that prediction into an experimental decision. This creates a clear scientific contribution, a measurable commercial benefit, and a demo that judges can understand in seconds. The strongest submission will show one real historical failure case, use leakage-safe compound-level evaluation, and clearly separate measured evidence from generated explanation.

Scientific constraint. The difference between two model probabilities is not automatically a ground-truth “translational gap.” The team should call it a *translational-discordance score* unless paired conventional-versus-human outcomes are available for supervision or calibration. This wording preserves the central idea without overstating what the data prove.

2 Problem, motivation and objective

2.1 The problem

Drug development relies on a sequence of representations of biology: molecular properties, simplified cell assays, animals, human-relevant in vitro systems, and finally patients. Failures arise when evidence from an early representation does not transfer to the next one. The issue is therefore more precise than “2D is bad and 3D is good”: early models can be scalable but biologically incomplete, while richer models can be informative but costly and low-throughput.

For the hackathon, the target failure mode is DILI. The operational question is:

Given a candidate drug, can LiverBridge detect a liability that a conventional molecular screen misses and recommend whether a human 3D liver assay is worth running?

2.2 Evidence motivating the project

The three supplied papers form a coherent argument:

1. **Human-relevant physiology improves translation.** Ewart et al. evaluated 870 Liver-Chips across a blinded set of 27 drugs and reported 87% sensitivity and 100% specificity for DILI, alongside an economic analysis of earlier failure detection [1].
2. **Information richness matters as well as geometry.** Bergen et al. introduced DILImap, comprising RNA-seq profiles for 300 compounds at multiple concentrations in primary human hepatocytes. ToxPredictor combined pathway-level toxicogenomic information with pharmacokinetics and achieved 88% sensitivity at 100% specificity in blind validation [2].
3. **Expensive biological experiments can supervise scalable virtual models.** Xu et al. developed ODFormer using over 14,000 response measurements across 183 pancreatic-cancer patient-derived organoids and 98 drugs, illustrating how a computational model can learn from organoid experiments rather than require a new physical organoid for every inference [3].

Together, these motivate a system that learns from human-relevant biology, estimates when simple screening is unreliable, and routes only the most informative compounds to expensive 3D testing.

2.3 Objective and success criterion

The MVP should transform one drug name or SMILES string into five outputs:

$$\text{drug} \rightarrow \begin{cases} \text{conventional molecular risk} \\ \text{human-biology DILI risk} \\ \text{discordance and uncertainty} \\ \text{mechanistic pathway evidence} \\ \text{3D validation recommendation} \end{cases}$$

The minimum convincing result is not a high aggregate score alone. It is one held-out or historical compound for which the system reproduces a meaningful translational discrepancy and gives an evidence-grounded recommendation.

3 Product concept and system architecture

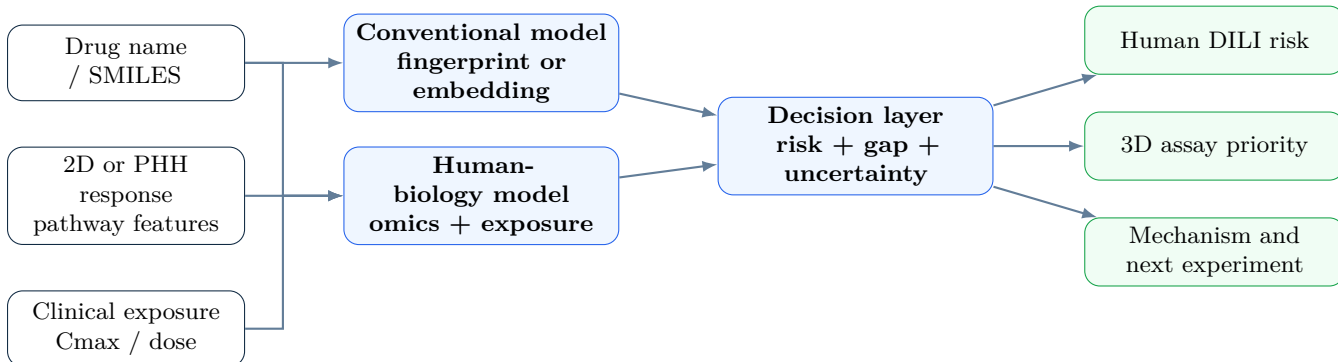


Figure 1: LiverBridge architecture. The scientific reasoning agent explains quantitative outputs and proposes validation experiments; it is not the toxicity classifier.

3.1 Recommended model stack

Component	Hackathon implementation	Purpose
Molecular baseline	Morgan fingerprint + logistic regression or XGBoost	Fast, interpretable reference for structure-only screening
Foundation variant	BioNeMo-compatible molecular embedding + small MLP	Demonstrates value of pretrained molecular representation
Human-biology model	Pathway response + exposure + molecular features; random forest or XGBoost	Estimates DILI using richer, human-relevant context
Uncertainty	Bootstrap ensemble, fold variance, or conformal set if feasible	Prevents overconfident triage and supports “test next” decisions
Reasoning layer	GPT-Rosalind supplied with structured model evidence	Explains pathways and proposes a discriminating experiment
Application	FastAPI backend + Streamlit or React frontend	Produces a judge-ready end-to-end demo

3.2 Decision rule

Let p_c be the calibrated structure-only DILI probability and p_h the calibrated human-biology probability. Define a prototype discordance score

$$D = p_h - p_c,$$

and an uncertainty score U from model disagreement or confidence intervals. A simple triage policy is:

$$\begin{aligned} \text{high priority : } & p_h \geq \tau_h \quad \text{and} \quad (D \geq \tau_D \text{ or } U \geq \tau_U), \\ \text{consider validation : } & p_h \text{ or } U \text{ is intermediate,} \\ \text{low priority : } & p_h < \tau_l \quad \text{and} \quad U < \tau_U. \end{aligned}$$

Thresholds should be set from validation data or explicitly labelled as demo thresholds. Do not present hand-selected thresholds as clinically validated.

4 Data and evaluation plan

4.1 Data table

Aim for one compound-level table with the following fields:

Field	Example	Role
Compound identity	Name, canonical SMILES, InChIKey	Deduplication and molecular encoding
Clinical outcome	DILI positive/negative or severity category	Primary target
Exposure	Cmax, tested concentration, safety margin	Separates activity at plausible exposure from extreme-dose effects
Biological response	Genes, pathways, or toxicogenomic signature	Human-biology input and mechanistic evidence
Conventional evidence	Animal, cytotoxicity, QSAR, or structure-only score	Comparator and translational-discordance context
Provenance	Source, assay, dose, replicate, confidence	Auditability and leakage control

4.2 Evaluation design

- **Split by compound, never by row.** All doses, replicates and measurements for one compound must remain in the same fold. Otherwise the model can recognise the drug rather than generalise to unseen chemistry.
- **Primary metrics.** AUROC, area under the precision-recall curve, sensitivity at a prespecified high specificity, specificity, Brier score, and calibration curve.
- **Decision metric.** Among compounds selected for 3D validation, report enrichment for clinical DILI positives or known preclinical false negatives.
- **Ablations.** Compare structure only, biology only, exposure only, and the combined model. The scientific claim is supported only if richer context adds information on held-out compounds.
- **Case studies.** Use one or more historical discrepancies such as troglitazone or published phase III failures, but label whether each case is held out, used in training, or included only as literature evidence.

Sample result format. “On unseen compounds, the combined model improved calibration and recovered a published human liver liability that the structure-only baseline ranked as low risk. Its uncertainty-aware policy would have prioritised that compound for Liver-Chip validation.”

5 Step-by-step build plan

1. **Lock the claim and interface (30 minutes).** Agree on DILI, one input contract, five outputs, and the language “decision support, not clinical validation.”
2. **Run a data feasibility gate (60–90 minutes).** Confirm that labels, structures and at least one human-biology or exposure feature set can be joined by compound. Count compounds after deduplication before building models.
3. **Create a leakage-safe dataset (2–3 hours).** Canonicalise identifiers, aggregate or retain dose appropriately, record provenance, and implement grouped train/validation/test splits.
4. **Train the reference baseline (1 hour).** Morgan fingerprints into logistic regression or XGBoost. Produce predictions, metrics and calibration plot.
5. **Train the richer model (2–4 hours).** Add pathway/toxicogenomic and PK features. Prefer a robust tree ensemble or small MLP over an ambitious architecture that cannot be validated.
6. **Compute triage outputs (1 hour).** Calibrate p_c and p_h , estimate uncertainty, calculate D , and implement a transparent recommendation rule.

7. **Build the evidence packet (1–2 hours).** For each prediction, export structured JSON containing probabilities, confidence, top pathways, data provenance and caveats.
8. **Add the scientific reasoning agent (1–2 hours).** Ask GPT-Rosalind to explain only the supplied evidence and propose the next experiment. Require citations or source identifiers in its response.
9. **Build the interface (2–3 hours).** Implement the ideal screen in Figure 2, the risk map in Figure 4, and a comparison view for several case studies.
10. **Rehearse and freeze (final 2 hours).** Cache demo compounds, test the offline path, verify every number, record a backup video, and keep a static result if live services fail.

5.1 Two feasible tracks

Track	When to use	Deliverable
A: Learned translator	Sufficient matched molecular, biological-response, exposure and DILI data are available	Quantitative model comparison, calibrated discordance, uncertainty-aware triage
B: Evidence-grounded prototype	Paired data cannot be assembled in time	Real structure-only baseline plus curated published 3D/toxicogenomic evidence for named case studies; clearly label unlearned components and avoid fabricated scores

Track B can still demonstrate product value, interface quality and agentic scientific reasoning. It must not masquerade a rule-based or curated score as a validated model.

6 Final demo and visualisations

6.1 Sketch of result screen

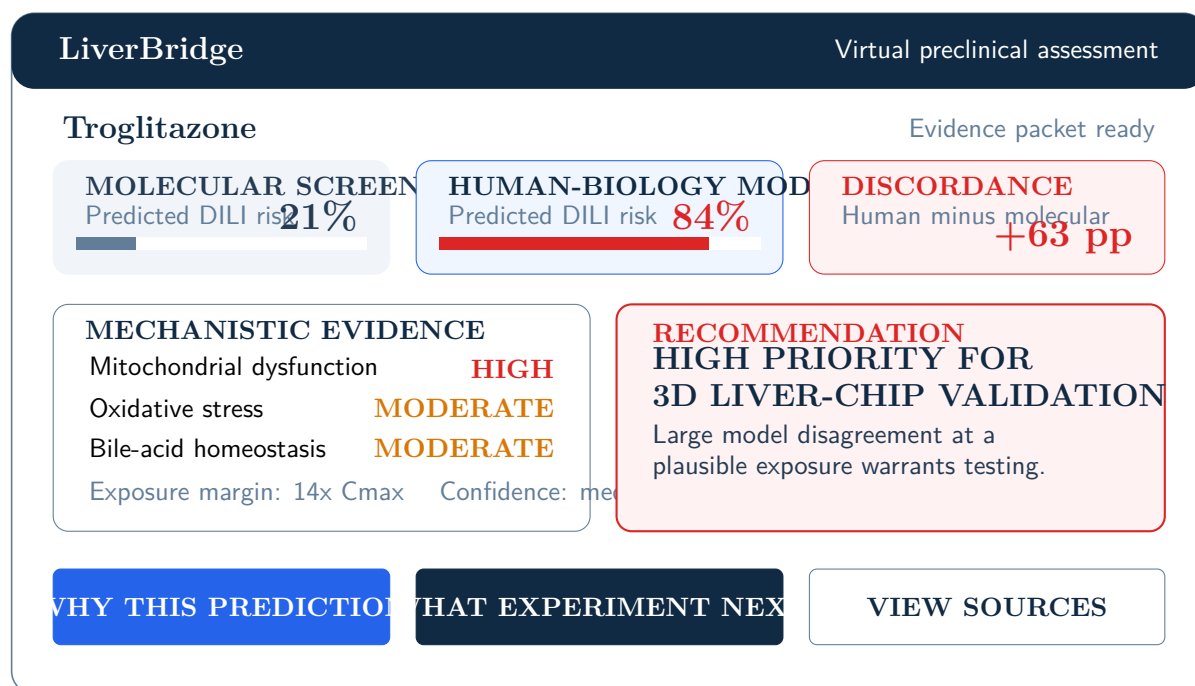


Figure 2: Ideal demo screen. Values shown are illustrative UI placeholders and must be replaced with model outputs or clearly identified literature evidence.

Candidate: Drug X

MOLECULAR SCREEN

Predicted DILI risk	18%
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HUMAN-BIOLOGY MODEL

Predicted DILI risk	81%
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TRANSLATIONAL GAP

+63%

⚠ Large disagreement with conventional prediction

CLINICAL EXPOSURE

Predicted toxic dose / Cmax	14×
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MECHANISMS

Mitochondrial dysfunction	↑↑↑
Oxidative stress	↑↑
Bile-acid dysregulation	↑

NEXT DECISION

HIGH PRIORITY FOR 3D LIVER-CHIP VALIDATION

WHY?

[Ask Rosalind]

WHAT EXPERIMENT NEXT?

[Generate suggested validation experiment]

Figure 3: Example of CLI screen.

6.2 Central visual: the translational-risk map

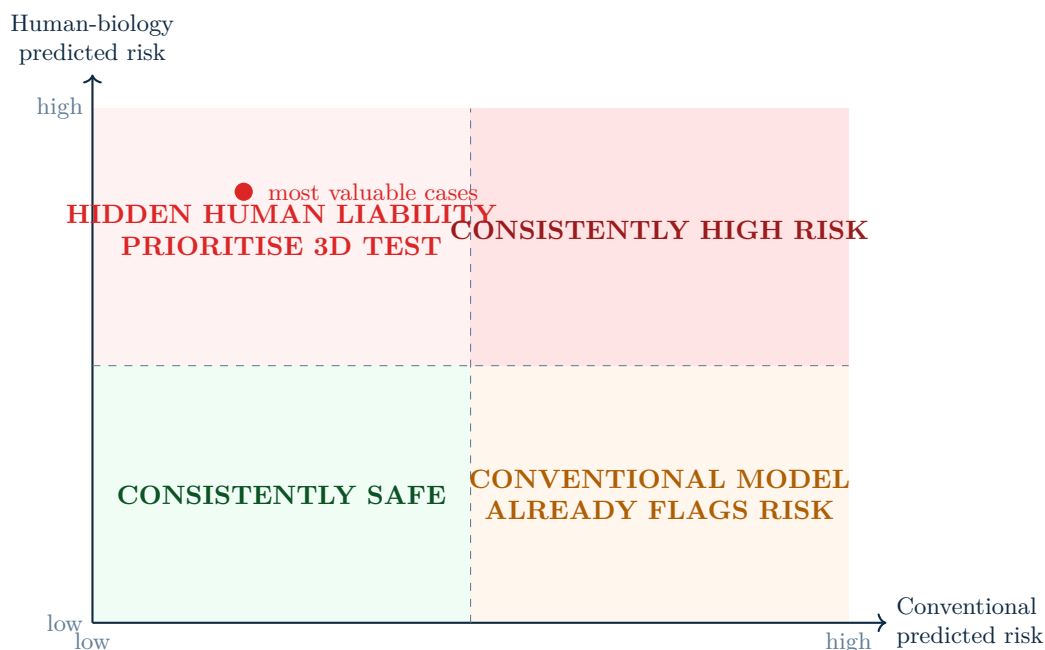


Figure 4: The upper-left quadrant is the product’s key target: compounds that appear relatively safe under a conventional model but risky when human biological context is introduced. Error bars or confidence regions should be shown in the interactive version.

6.3 Commercial visual: selective escalation funnel

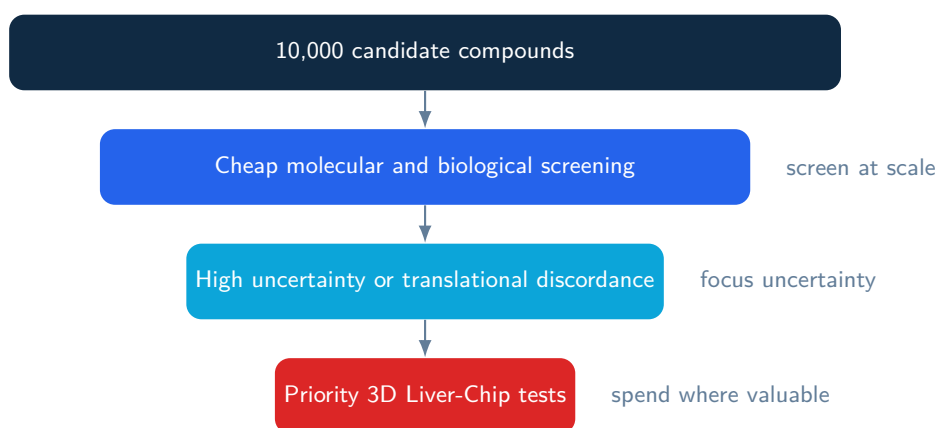


Figure 5: LiverBridge does not promise to replace organoids or organs-on-chips. It aims to allocate them efficiently. Counts after the first stage are illustrative and should be replaced with observed selection rates.

7 Team execution and tool allocation

Workstream	Owner profile	Concrete output
Biology and claims	Toxicology / drug discovery lead	Label definition, valid assay interpretation, pathway review, case-study selection
Data and modelling	ML lead	Compound table, grouped splits, baselines, combined model, uncertainty and evaluation
Molecular representation	BioNeMo / cheminformatics lead	Canonical structures, fingerprints or cached foundation embeddings
Scientific agent	GPT-Rosalind lead	Evidence-constrained explanation schema and next-experiment prompt
Product and demo	Full-stack lead with Codex	API, visual interface, cached examples, source panel and backup recording
Pitch and validation	Team lead	Three-minute narrative, quantitative claims audit, limitations and business case

Suggested infrastructure. Use Brev for a reproducible GPU environment if foundation embeddings require it. Use Codex for data plumbing, tests, API integration and UI generation. Cache embeddings and demo responses so the final presentation does not depend on network latency.

8 Risks, safeguards and non-goals

Risk	Why it matters	Mitigation
No matched 2D/3D dataset	Direct translation cannot be learned or evaluated	Use Track B; call output discordance, not ground-truth gap
Compound leakage	Inflates performance through repeated doses or replicates	Group all rows by compound before splitting
Small, imbalanced data	AUROC may look good despite poor positive prediction	Report AUPRC, calibration and confidence intervals
Uncalibrated probabilities	Difference between risk scores becomes meaningless	Calibrate both models on held-out validation data
LLM hallucination	Mechanistic story may outrun evidence	Restrict agent to structured features and cited sources
Clinical overclaim	Prototype could be interpreted as validated safety advice	Label as research decision support; disclose dataset scope
Live-service failure	Breaks the final demonstration	Cache three case studies and record a backup walkthrough

The prototype is **not** a virtual human, a mechanistic whole-organ simulator, a regulatory-grade predictor, or a replacement for 3D experiments. Its credible claim is narrower: prioritising where human-relevant testing may add the most value.

9 Pitch structure and judging checklist

9.1 Three-minute pitch

- Problem (20 s):** cheap preclinical models scale but miss human context; richer 3D assays are too expensive to run everywhere.
- Insight (20 s):** predict the failure of the model, not only the failure of the drug.
- Product (30 s):** LiverBridge compares conventional and human-biology predictions, quantifies discordance and recommends 3D validation.
- Live case (70 s):** enter a drug, show the risk map, mechanisms, uncertainty and next experiment.
- Evidence (25 s):** report held-out metrics, ablations and one historical translational failure.
- Impact (15 s):** focus expensive human-relevant experiments on the candidates where they can prevent costly late failure.

9.2 Definition of done

- One command launches the application; three cached compounds always work.
- Every displayed number is traceable to model output or a cited source.
- Train/test splitting is grouped by compound and described on one slide.
- At least one baseline and one combined model are compared on the same held-out set.
- Uncertainty and limitations are visible, not buried in narration.
- The agent explains evidence; it does not invent a toxicity score.
- The demo ends with a concrete experimental recommendation.

Closing line. LiverBridge turns expensive human-relevant biology from a bottleneck into supervision: learn where conventional screens break, then deploy 3D experiments only where they can change the decision.

References

- [1] L. Ewart et al., “Performance assessment and economic analysis of a human Liver-Chip for predictive toxicology,” *Communications Medicine*, vol. 2, article 154, 2022. [doi:10.1038/s43856-022-00209-1](https://doi.org/10.1038/s43856-022-00209-1).
- [2] V. Bergen et al., “A large-scale human toxicogenomics resource for drug-induced liver injury prediction,” *Nature Communications*, vol. 16, article 9860, 2025. [doi:10.1038/s41467-025-65690-3](https://doi.org/10.1038/s41467-025-65690-3).
- [3] J. Xu et al., “ODFormer: a Virtual Organoid for Predicting Personalized Therapeutic Responses in Pancreatic Cancer,” bioRxiv preprint, 2025. [doi:10.1101/2025.07.08.663664](https://doi.org/10.1101/2025.07.08.663664).